Foldable acoustic metamaterial for underwater broadband low-frequency noise mitigation

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(Received 3 September 2025; revised 26 November 2025; accepted 16 December 2025; published 9 February 2026)

This paper is a contribution to the Physical Review Applied collection titled Phononics and Metamaterials.

Underwater noise control has become increasingly important recently due to its tremendous impact on marine ecosystems. The various proposed solutions have been hampered by their insufficient noise reduction level, low-frequency wideband inefficiency, or transportation and deployment inconvenience. In this work, we present an acoustic metamaterial targeted at ultrawideband low-frequency noise attenuation in underwater environments. The design employs air-entrained channels to establish waveguides with soft boundaries that suppress acoustic wave propagation below a designed cutoff frequency. This mechanism enables effective low-frequency sound insulation over a broad bandwidth with a deep subwavelength scale, and employs a lightweight and foldable structure. Underwater experiments confirmed our metamaterial prototype can achieve an average insertion loss of 29.108 dB and a maximum noise reduction of 60.362 dB across the predicted band gap from 1.5 to 15 kHz over more than three octaves. This design has demonstrated its great potential for large-scale applications, such as offshore wind-farm construction, bridge piling, and deep-sea oil operations.

DOI: 10.1103/3j5q-66c3

I. INTRODUCTION

Underwater noise control is essential due to its critical ecological impacts on marine life and sustainable development [1–7]. Large-scale offshore facilities are the primary sources of underwater noise [8–10]. Significant low-frequency noise generated during the construction, operation, and demolition phases can be difficult to mitigate and may propagate over distances reaching several hundred kilometers.

Existing commercialized noise mitigation systems such as bubble curtains, isolation castings, hydrosound dampers, and underwater Helmholtz resonator arrays have been maturely studied and extensively deployed to reduce radiation noise [11]. Nevertheless, these aforementioned conventional solutions often suffer from high installation and/or operational costs, limited bandwidth, low-frequency ineffectiveness, and large-scale deployment inconvenience [12,13].

Targeting these limitations, recent publications on underwater acoustic metamaterials have highlighted their compact and tunable solutions for effective noise control. By engineering the fundamental building blocks, or meta-atoms, one can realize behaviors that are otherwise difficult with traditional structures. Previous researchers have, in general, classified various metamaterial designs into two functional classes: sound absorption and sound insulation metamaterials [14–16].

Absorber-type metamaterials rely on establishing an impedance-matching layer with strong energy dissipation capability. They can be further divided into two subcategories: local-resonance-based and nonlocal-resonance-based realizations [15]. Local-resonance-based absorbers exploit the resonant modes of various subwavelength structures to enhance energy dissipation. Representative designs include locally resonant phononic crystals [17–19], air bubble arrays [20,21], honeycomb quasi-Helmholtz resonators [22], thickness-graded elastic plate scatterers [23], and other composite-ingredient metasurfaces [24,25]. These designs demonstrate remarkable absorption performance, with coefficients exceeding 0.9 and in some instances approaching unity. However, the absorption spectra are usually composed of discrete bands, imposing intrinsic constraints on broadband performance, consistency, and tunability.

On the other hand, nonlocal-resonance-based absorbers utilize thermoviscous dissipation and sound-localization mechanisms. For instance, in the metastructure developed by Gao and Zhang [26], a helical metal embedded with viscoelastic rubber facilitates the destruction of the incident longitudinal wave and enhances shear-wave deformation of the damping material. Similar ideas can be found in Yu *et al.*'s design using metallic plate insertion [27], Lee *et al.*'s design of a finlike metal surface [28], and Dong *et al.*'s porous-solid metaconverters [29]. Furthermore, by controlling the local refractive index, exponential-gradient

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absorbers can concentrate acoustic energy to its interior region while minimizing transmission [30]. Despite their versatility, nonlocal resonance absorbers are shown to be more effective at relatively high frequency, and practical deployment remains challenged by manufacturing complexity and limited experimental validation.

Insulation-type metamaterials, in contrast, strive to establish a tremendous impedance mismatch between water and designed structures, with the aim to enhance reflection and reduce transmission [14]. Such metamaterials are often used as decoupling layers, which can acoustically separate the underwater vibrating structures from the surrounding water environment [31]. Porous metamaterials and microvoid elastomers are classical choices for the insulation layer. This class of insulators is usually designed for plate coatings. Previous researchers have proposed various cavity-entrained structures and employed the equivalent fluid model, vibroacoustic coupled model, and topology optimization to develop mathematical characterization [31–33]. Even though they have demonstrated some abilities in vibration control at any desired frequency, problems like low-frequency boost below ∼2 kHz and manufacturing infeasibility still greatly limit the potential of such prototypes. Local resonators also appear in the insulation-type designs. At resonant frequencies, an abrupt change in complex effective density or bulk modulus can lead to boosted reflection. Examples include periodically perforated rubber layers and ultrathin composite plates [34–36]. However, successful fabrication remains a tremendous challenge, and even those structures with experimental validations could not guarantee a steady broadband noise reduction capability, especially for frequencies below 1–3 kHz.

In this paper, we present an underwater acoustic metamaterial with an air-entrained-based structure at deep-subwavelength scale. Theoretical design showed agreement with supporting simulations and experimental results. Our lab-scale prototypes have firmly demonstrated an excellent ultrabroadband noise attenuation ability of 30 dB across more than three octaves in the low-frequency regime (1.5–15 kHz) with a maximum performance of ∼60 dB. Furthermore, the foldability and ease of manufacturing of our design can facilitate low-cost storage, transportation, and deployment, paving the way for commercial noise mitigation systems. Continuing from this introduction, Sec. II will be focused on metamaterial design and realization. Based on solid simulation results in Sec. III, Sec. IV will highlight fabrication and experimental details.

II. METAMATERIAL DESIGN AND REALIZATION

This study models the problem as illustrated in Fig. 1(a). A sound source radiates broadband, intense low-frequency noise into the surrounding ocean or lake environment. The blue curtain depicts the appearance of the desired metasurface, which can be extended and shield noise radiation paths from all directions.

A. Meta-atom underlying principles

To allow unobstructed water flow around the noise source, the design is developed based on the structure of an underwater acoustic waveguide. The water can then travel through the metasurface freely. Figure 1(b) shows an enlargement of a local section of the metasurface consisting of 13 small waveguides, and each waveguide, as depicted in Fig. 1(c), is composed of four sidewalls. The waveguide walls are designed to mimic soft (pressure-release) boundaries, leading to a purely imaginary wave number in the forward-propagating direction below its first cutoff frequency. Such a cutoff mechanism allows us to achieve efficient noise mitigation at low frequencies using subwavelength structures.

Consider a two-dimensional rectangular waveguide model [Fig. 1(d)] in the x-y plane that allows acoustic waves to propagate along the z direction unboundedly. Fundamental acoustics describe that, for an ideal and uniform fluid medium, the three-dimensional (3D) wave equation takes the following form:

$$\nabla^2 p = \frac{1}{c_0}\frac{\partial^2 p}{\partial^2 t}, \tag{1}$$

where p is the acoustic pressure and c_0 is the speed of sound of the background medium (water) in the equilibrium state. Assuming the time-harmonic solution of the wave equation

$$p(x,y,z,t) = \mathrm{Re}\{\hat{p}(x,y,z)e^{-i\omega t}\},$$

one can then obtain the homogeneous acoustic Helmholtz equation as $(\nabla^2 + k^2)\hat{p} = 0$, where $k = \omega/c_0$ denotes the wave number of the background medium. Further simplification yields the relationship among wave numbers in all three directions:

$$k = \sqrt{k_x^2 + k_y^2 + k_z^2}. \tag{2}$$

Now, by imposing the pressure-release boundary conditions on the x-y plane, namely $\hat{p}_{x=a/2} = \hat{p}_{x=-a/2} = \hat{p}_{y=b/2} = \hat{p}_{y=-b/2} = 0$, the space-varying acoustic field can be solved as

$$\hat{p}(x,y,z) = \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{\pm jk_z z}, \tag{3}$$

where $m,n = 1,2,3,\ldots$ are positive integers, and every combination of (m,n) stands for an independent mode. Parameters a and b are the width and height of any particular rectangular waveguide, respectively. Hereafter,

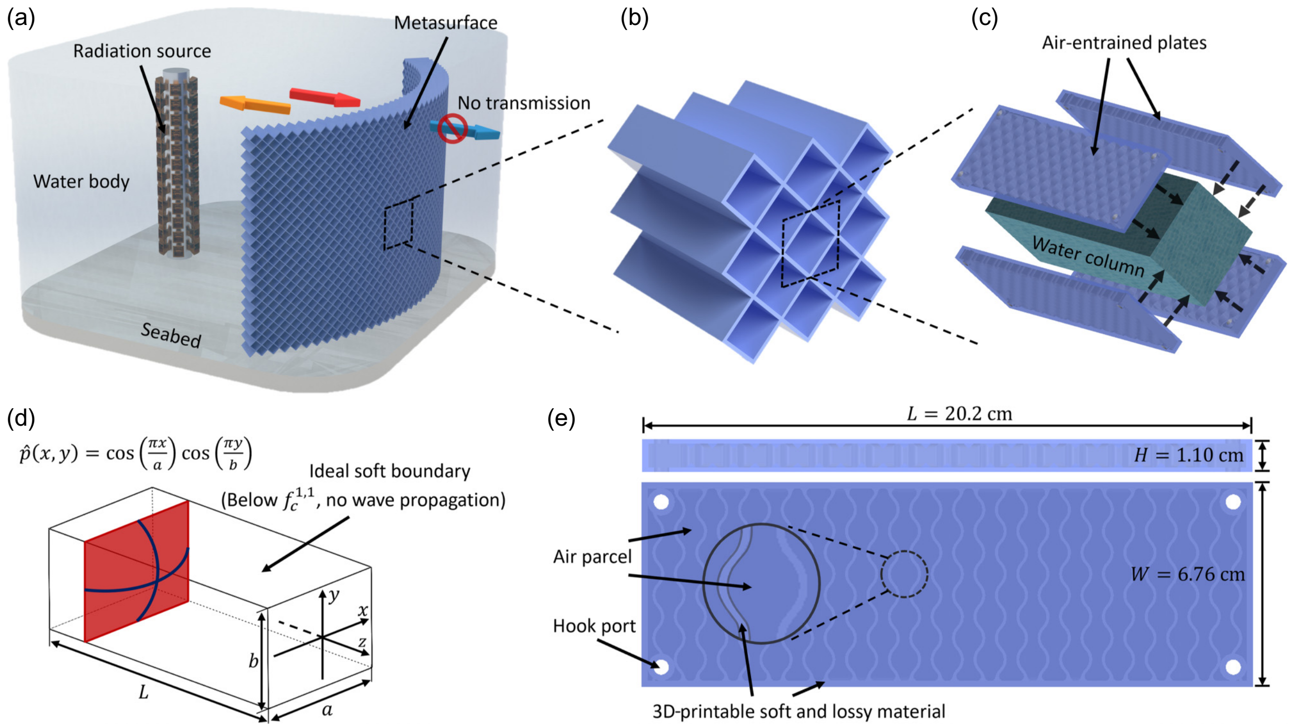

FIG. 1. Air-entrained plate design guidance and specification. (a) Problem modeling of underwater acoustic mitigation via insulation-based metasurface. (b) A selected section of the metasurface consisting of 13 acoustic waveguides. (c) The fundamental element of the metasurface referred to as a meta-atom. This shows a disassembled view, and black arrows indicate the attachment of plates to the water column. (d) Sketch of acoustic waveguide theory with soft boundary. Red segments indicate the spatial distributions of the pressure field. (e) A single as-designed air-entrained plate in top and front views.

the wave number in the propagating direction can be expressed as

$$k_z = \sqrt{\left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}. \tag{4}$$

Observe that, as

$$\left(\frac{\omega}{c}\right)^2 < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2, \tag{5}$$

or equivalently

$$f < \sqrt{\left(\frac{mc}{2a}\right)^2 + \left(\frac{nc}{2b}\right)^2} \equiv f_c^{\,m,n}, \tag{6}$$

k_z takes purely imaginary values. Here, f is the frequency in hertz and f_c is defined as the cutoff frequency. This relation indicates that, for every single mode, waves with a frequency below the cutoff bound become evanescent. Additionally, its imaginary part contributes to the decay rate in the background medium, and a longer waveguide showcases more prominent noise attenuation ability. This unique property of an acoustic waveguide with soft boundaries can lead to a super-wide band gap within $f \in (0, f_c]$, which favors sound mitigation, especially for low frequencies.

For the desired subwavelength metamaterial design targeting low-frequency sound, only the lowest mode $(m, n) = (1, 1)$ of the propagating waves is of interest, as no wave is allowed to propagate below $f_c^{\,1,1}$. Tuning the geometry of the waveguide can yield fully customized noise mitigation behaviors. As an example, for a square-cross-section waveguide with $a = 7$ cm, the designed cutoff frequency is about 15.15 kHz [Fig. 2(a), "Ideal Soft-Boundary Waveguide Mode"].

Downsizing in the waveguide dimension is beneficial for reaching a broader band gap. In the limiting case where the lattice space is entirely filled with soft-boundary domains, the band gap becomes effectively infinite. However, such a highly compact configuration deviates from practical design considerations as it can hinder water permeability through the metasurface, making it less tolerant to ocean currents. Additionally, the denser structure increases the overall weight and poses greater challenges for larger-scale fabrication, transportation, and deployment. Therefore, a reasonable balance between the waveguide size and the desired acoustic performance must be sought to ensure both broadband attenuation capability and practical deployability.

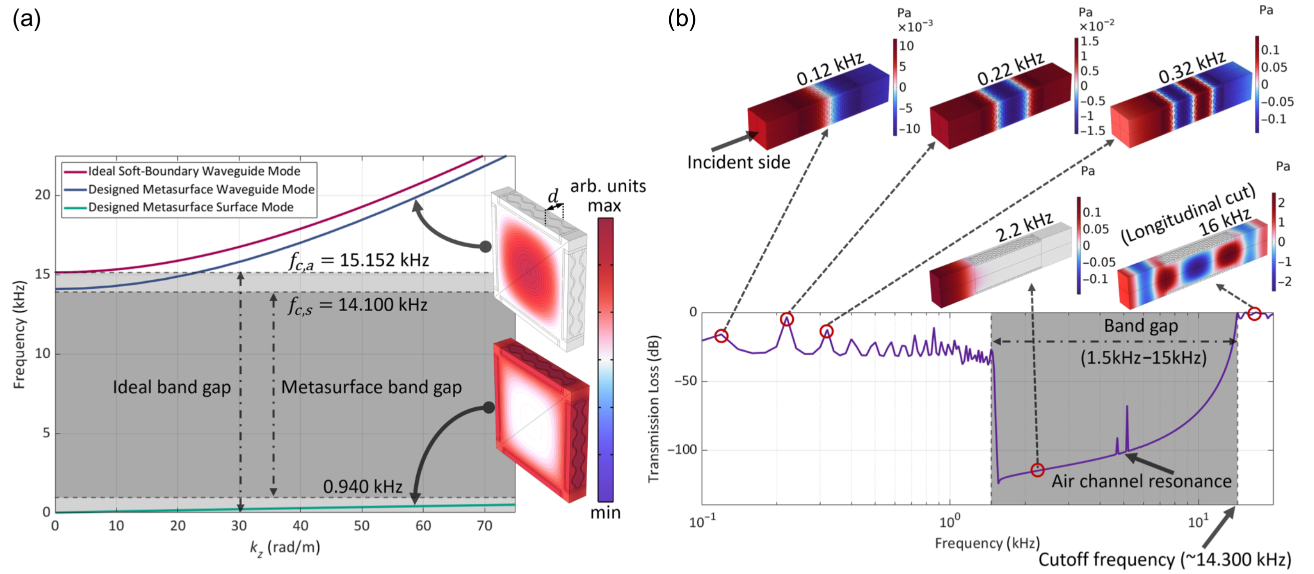

FIG. 2. (a) Band-structure diagram from COMSOL simulation along with two representative unit sections, both having the same Floquet periodicity d. The top unit section corresponds to the expected mode through water, while the bottom unit section represents the plate wave mode. The colored profiles visualize the acoustic pressure fields of the representative modes, with the color scale bar on top indicating the corresponding total acoustic pressure. (b) Frequency-domain simulation response of metasurface design with thickness $L = 20.2$ cm. Each waveguide on the top illustrates the total acoustic pressure field at their corresponding frequencies with the front face as the incident side. The waveguides at 2.2 and 16 kHz are longitudinal profiles revealing the wave propagation in the water column. The accompanying color bars provide a quantitative reference for the total pressure magnitudes shown in these modal fields.

B. Metasurface realization and band-structure analysis

To realize the soft boundaries, air serves as an ideal candidate, as its characteristic impedance (415 Rayl) is significantly lower than that of water (1.5×10^6 Rayl). However, since air bubbles in the water medium are extremely unstable, we use air-entrained plates as the containers of air parcels to approximate the pressure-release sidewall.

However, the introduction of external supporting structures can lead to unfavorable shear-related modes that allow waves to propagate through solids. They are typically the plate shell's inherent vibrations as a result of solid-fluid interactions. To minimize shear-wave coupling, we employ 3D-printable flexible resin materials with a shear modulus significantly lower than their Young's modulus. The internal structure of the air-entrained plate is designed as wavy supports to enhance its mechanical stability and withstand hydrostatic pressure in the water environment, while ensuring suitability for 3D printing [Fig. 1(e)]. Four plates with shared edges then form a single waveguide unit.

To further investigate and validate the band structure, an eigenmode analysis is performed in COMSOL Multiphysics 6.2. A unit section of the waveguide with a width of d was selected (which covers exactly two wavy air columns), as shown in Fig. 2(a). Floquet periodicity with the forward-propagating wave number k_z sweeping from 0 to π/d was applied to the boundaries. The band structure in Fig. 2(a) shows good agreement between simulation and theoretical calculation of the waveguide modes, indicating that the entrained air cavity is a close approximation to soft boundaries at higher frequencies. In addition to the expected band from the metasurface waveguide mode, there also exists another band occupying a much lower frequency regime. The bottommost curve in Fig. 2(a) outlines this phenomenon, corresponding to a mode arising from physical realization. This band allows extremely low-frequency wave components below 0.940 kHz to propagate, impairing the noise insulation at extremely low frequencies. The mode shape of this lower band shows that this extra band is attributed to wave propagation through air-entrained plates. Sound propagation is very slow (~39.6 m/s) and highly contained inside the plates, making such a mode very sensitive to losses within the plate materials. Therefore, materials with high loss (i.e., high hysteresis) are beneficial for damping out these plate waves quickly. Altogether, based on the need for low-shear and lossy materials, we opted to use the 3D-printable SuperElastic 60A Soft Silicone from 3DMaterials, which has measured speed of sound, loss factor, density, and shear modulus of $c = 1871.2$ m/s, $\delta = \mathrm{Re}\{k_z\}/\mathrm{Im}\{k_z\} = 0.0326$, $\rho = 1181.9$ kg/m^3, and $G = 1.534$ MPa.

Moreover, in the structural realization, the finite-size air cavity can no longer be considered as a perfect soft boundary; therefore, an additional acoustic impedance analysis is performed to quantitatively evaluate the deviation. The normal incidence of a plane-wave pressure onto a single air-entrained plate is considered and characterized as the infinite repetition of a single air parcel in the COMSOL simulation [Fig. 3(a)]. The spatially averaged pressure $p(f)$ and particle velocity in the normal direction $v_n(f)$ of the impact-plate domain are evaluated, appearing in the normalized acoustic surface impedance calculation:

$$z(f) = \frac{Z_s(f)}{Z_w} = \frac{p(f)/v_n(f)}{Z_w}, \tag{7}$$

where $Z_w = 1.50 \times 10^6$ Rayl is the characteristic impedance of water. Figure 3(b) demonstrates the frequency response of the surface impedance from the simulation. As illustrated, the real part stays extremely close to zero while the imaginary part crosses over zero at 1086.6 Hz. This behavior closely resembles that of a classical lumped spring-mass system, indicating the oscillatory property of the air-entrained channels. However, because the imaginary part is within the range of $(-0.04, 0.08)$ for the frequency of interest, it is still a reasonable quasiperfect soft boundary, which ideally possesses zero imaginary part.

Besides the designed noise attenuation mechanism through the waveguide theory, air cavity resonance can theoretically also contribute to this target. From the previous analysis of surface impedance [Fig. 3(b)], the imaginary part becomes zero at 1086.6 Hz, which corresponds to the resonance behavior of the air channels. However, this frequency has been embedded in the plate wave modes as illustrated in both the band-structure diagram or the frequency-domain simulation in Fig. 2. Hence, controlling this resonance and its higher-order modes are outside of the main consideration of this study. More essentially, such vibration has almost negligible noise insulation level. Considering an infinitely long air-entrained plate, the vibroacoustic simulation finds its lowest resonance frequency at $2892.5 + 4007i$ Hz. This low-Q resonance can provide very limited noise reduction, as it is masked out by the highly effective band gap. Hence, even though such vibration is observable, it does not contribute much to the overall spectrum of the system.

Finally, the metasurface is designed as a foldable structure for easy transportation and deployment. Each plate from a single unit cell is loosely connected by zip ties through the hook ports on the corners to allow a rotational degree of freedom [Fig. 1(e)]. The metasurface as a whole then resembles an accordion gate, and tessellated square cells guarantee smooth morphing when setting up the structure at the scene. The structure will remain compactly folded until the actual installation in the underwater environment. Ideally, due to the net effect of buoyancy, gravity, and supplementary tension, the metasurface will be completely unfolded and act as a shielding layer. To accommodate the size of our available testing water tank, the single plate was designed with dimensions of $L = 20.2$ cm, $W = 6.76$ cm, and $H = 11.0$ cm [Fig. 1(e)].

III. NUMERICAL SIMULATION

In COMSOL Multiphysics, we performed a frequency-domain simulation on a single meta-atom geometry with symmetry boundaries to mimic an infinitely large metasurface. Both ends of the meta-atom are topped with water columns, and plane-wave radiation is applied to the incident and transmitted boundaries, with 1-Pa incident pressure from the left [Fig. 2(b)]. The transmission loss (TL) is then calculated as $10 \log_{10}(W_t/W_i)$, the quotient of transmitted by incident sound power in the decibel scale.

Figure 2(b) shows the transmission loss of the designed structure from 100 to 20 000 Hz. The frequency-domain simulation demonstrated a good agreement with the band-structure analysis from Fig. 2(a). A forbidden band with high transmission loss can be observed from roughly 1.5 to 15 kHz. Within this band gap, the structure exhibits nearly zero transmission, effectively achieving the goal of noise mitigation in a wide frequency range of more than three octaves. A field plot at 2.2 kHz confirmed the evanescent behavior of the wave inside the waveguide. Right above the cutoff frequency, such as at 16 kHz, the waveguide exhibits a transmission mode through the water content, which corresponds to the the $(1, 1)$ waveguide mode in Fig. 2(a).

Below a frequency of around 1.5 kHz, the response shows a series of resonant behaviors with a comparably low transmission loss. As elucidated in Sec. II, the flat response is mainly due to the plate wave modes; at some selected air channels' resonance frequencies of 0.12, 0.22, and 0.32 kHz, the total acoustic pressure distribution on the plates indeed characterizes the plate transmission.

To further investigate the performance limits of our design, additional frequency-domain simulations were conducted with various metasurface thickness (i.e., the length of air-entrained plate). The results indicate that increased longitudinal dimension provides enhanced noise attenuation ability within the predicted band gap while having extremely limited influence on the plate wave modes. A more detailed analysis is provided in Appendix A.

IV. FABRICATION AND EXPERIMENT

To validate the performance of our designed metamaterial, we carried out two sets of underwater experiments. The first configuration, the metasurface screen, verified the design but suffered from a source-tank coupling effect that reduced low-frequency accuracy. To address this issue, a second configuration, the metasurface cage, was developed

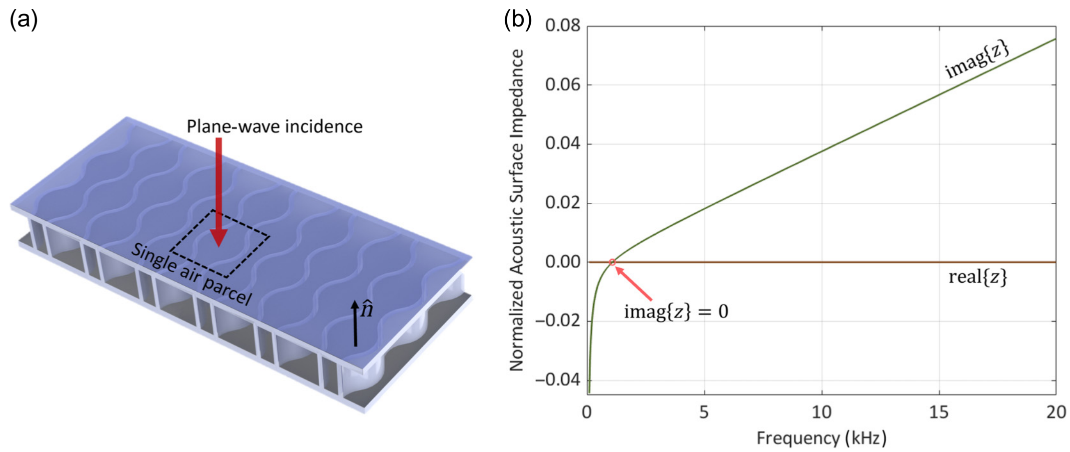

FIG. 3. Acoustic surface impedance calculation. (a) Cross section of a single air-entrained plate, where the hard-bounded domain of a single air parcel is used as a repetition unit. (b) Frequency response of normalized acoustic surface impedance, where the brown and teal curves represent the real and imaginary parts of the surface impedance.

to further decouple the source from the tank and obtain more reliable experimental results.

A. Metasurface screen

The proposed air-entrained plates are fabricated using ELEGOO Saturn 3 Ultra 3D printers with SuperElastic 60A Resin from 3DMaterials. To test our metasurface in a $1.22 \times 0.53 \times 0.45\ \mathrm{m}^3$ water tank, we assembled a metasurface screen with a 3-by-4 meta-atom diagonal arrangement, yielding a total of 18 complete and 10 half-complete waveguides [Fig. 4(a)]. The plates are all connected with zip ties, and each interior joint can accommodate four sides. The metasurface is mounted on linear rails to maintain its general shape. This frame structure also stabilizes the metasurface in place for underwater testing without the need for extra anchoring weights. As expected, by sliding the rails, the metasurface screen can be easily folded, forming a lightweight, flexible structure for low-cost transport [Fig. 4(a)].

The experiment was conducted in a water tank filled with de-ionized water. As illustrated in the experimental schematic in Fig. 4(b), a computer sends single-frequency sine signals via MATLAB to the data acquisition unit (DAQ, National Instruments USB-6363). The output signal is then directed to a sound source on the incident side after passing through an audio amplifier (Fosi Audio TB10A TPA3116). The sound source consists of a surface speaker (Dayton Audio Thruster EX32EP2-4) fastened to a 3D-printed shell, with a solid plate acting as a vibrating panel placed in contact with the speaker and covered by a polydimethylsiloxane (PDMS) membrane for waterproofing. On the transmission side, a hydrophone (Aquarian Scientific AS-1) with a quasi-uniform free field receiving sensitivity of -208 dBV re 1 μPa, mounted on an aluminum rod and positioned by servomotors, captures the transmitted pressure signal and sends it back to the DAQ as input. Finally, to reduce undesired reflections from the glass walls, two clay-coated acrylic plates are placed at both ends of the tank, with the speaker mounted on the plate at the incident side.

We performed two measurements: one with and one without the designed metasurface screen. For each frequency $f_i \in [100, 20\,000]$ Hz with a fixed step size of 20 Hz, a 3-s sine wave was generated and used as the input signal. The first 0.5 s of the received signal was discarded to only capture the steady-state response. Then the time-domain signal is Fourier-transformed to obtain its amplitude at each frequency. Figure 4(c) illustrates the sound pressure level (SPL) for both cases along with their SPL difference,

$$\Delta\mathrm{SPL}(f) = \mathrm{SPL}_1(f) - \mathrm{SPL}_2(f), \tag{8}$$

where the subscripts 1 and 2 denote the cases without and with the metasurface, respectively. According to the ideal waveguide simulation, the designed metasurface opens a band gap in the frequency regime of approximately $f \in [1500, 15\,000]$ Hz, which is shaded in gray. It is noted here that, since the measured wavelength ($\sim$1.5 m at 1000 Hz) is comparable to the size of the water tank (1.22 m), and that an anechoic boundary is not feasible in experiments, our experiments show insertion loss rather than transmission loss. Nevertheless, the measured noise reduction still achieves good agreement with our calculations and simulations, showing significant noise reduction within the band gap. Our metasurface screen achieves an average insertion loss of -20.701 dB within this theoretical band gap, reaching up to -32.007 dB across two octave bands. At 7.26 kHz, a maximum effectiveness is attained with a value of -66.793 dB. For even higher frequencies outside the band gap, this structure is still functional at some selected

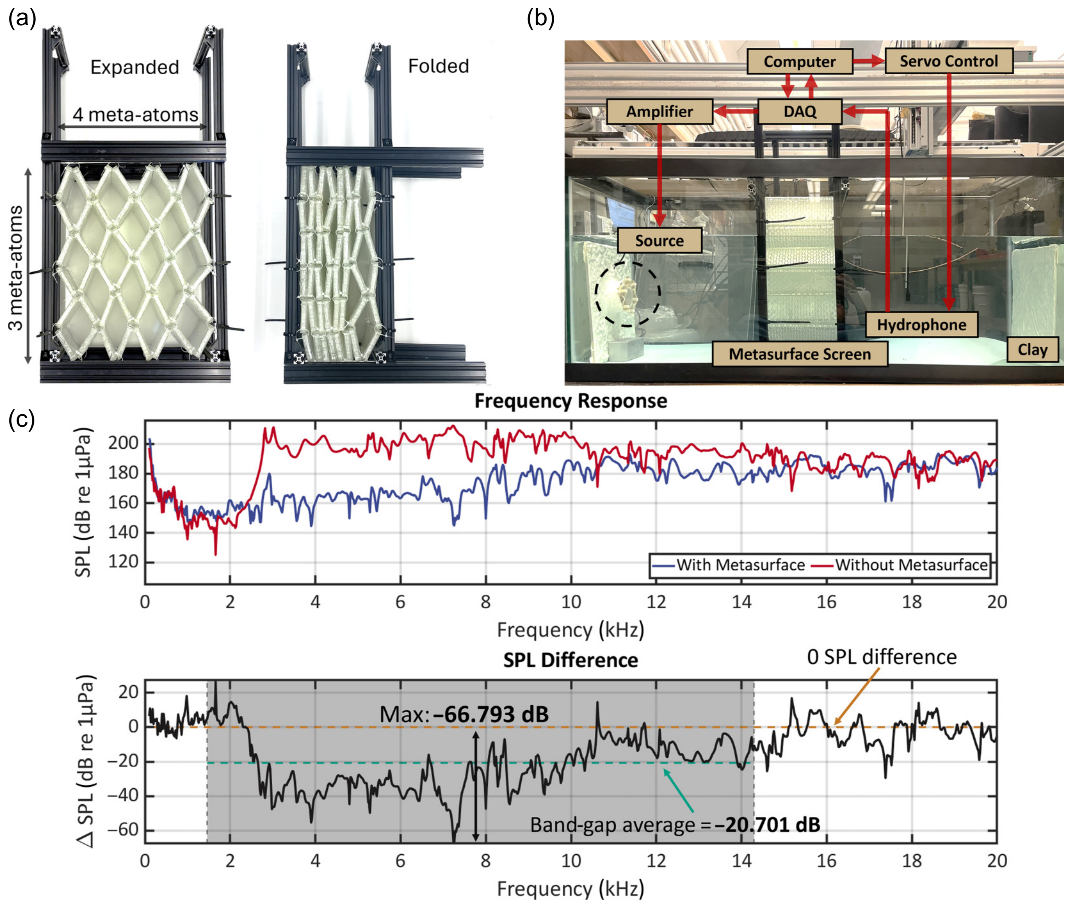

FIG. 4. Experiment I—metasurface screen. (a) Fabricated prototype consisting of a 3-by-4 diagonally arranged meta-atom array in the fully expanded shape (left) and folded shape (right). (b) Experimental setup. (c) Sound pressure level (SPL) frequency responses of tests with and without the metasurface screen (top) and SPL difference characterizing the effectiveness of the design (bottom).

frequencies, but a general trend of oscillating around the reference level (zero SPL difference) can be observed.

However, at the very beginning of the band gap and even lower frequencies, the SPL difference is low. While plate wave modes and nonideal soft boundaries can contribute to the observed phenomenon, the low ΔSPL mainly results from the inherent resonance of the testing water tank, and the sound is transmitted through the vibrating glass plates. Since the sound source directly lies against the glass wall, it can excite the water tank's vibrational modes at various frequencies, especially at frequencies below 2300 kHz. Hence, such vibrations can propagate to the transmission side and be picked up by the hydrophone, regardless of what structure is put in the water domain. A more detailed analysis of these vibrations is provided in Appendix B.

B. Metasurface cage

As noted, the measurement accuracy at extremely low frequencies is limited by the water tank's inherent modes. To avoid this limitation, we built an acoustic metasurface cage that can enclose the sound source and physically isolate the speaker from the tank walls.

As shown in Fig. 5(a), the cage consists of four vertical screens, each formed by a 2.5-by-2 meta-atom arrangement serving as the sides, and one cross-shaped meta-atom forming the bottom. To accommodate the dimensions of the water tank, the length of each air-entrained plate was resized to $L' = 10.3$ cm, while the width, height, and material (SuperElastic 60A Resin) remained unchanged. In the previous experiment, some plates on the margins were torn because their fragile hook ports bear excessive tension. To address this issue, we 3D-printed frames with hook ports using ELEGOO Standard Photopolymer Resin to secure the plates, as the higher stiffness of the material can better withstand tensile stress [Fig. 5(a)]. Similarly, plates within each screen are zip-tied together, and the screens are attached to square rails at both ends to form a complete cage. Four anchoring weights are deployed to stabilize the cage in the tank [Fig. 5(b)]. We assembled another speaker in the same fashion and connected two speakers face-to-face to mimic an axial symmetric sound source. The sound source is suspended in the water tank by a fishing line.

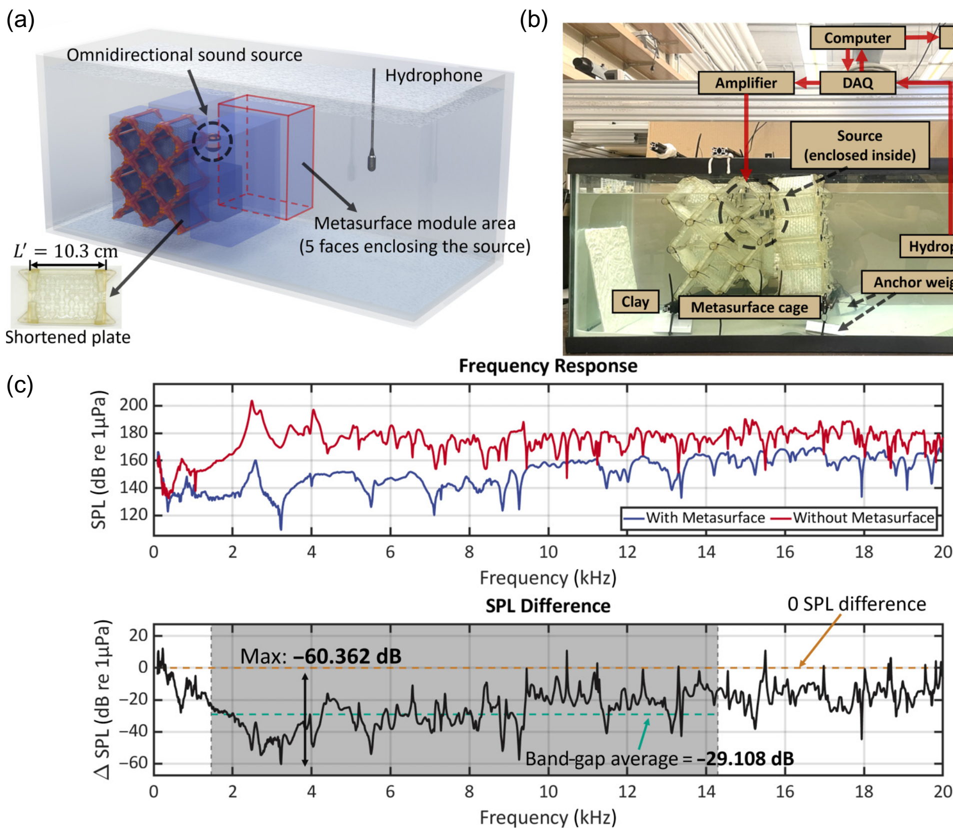

FIG. 5. Experiment II—metasurface cage. (a) Illustration of metasurface cage assembly. Each blue region (four on the side and one on the bottom) represents the placement of a metasurface screen. The side screen is arranged in a 2.5-by-2 configuration, while the bottom one is simply a unit waveguide. This cage also encloses an omnidirectional sound source composed of two surface speakers facing each other. (b) Experimental setup. The enclosed sound source is circled in black. (c) SPL frequency responses of measurements with and without the metasurface screen (top) and SPL difference characterizing the effectiveness of the design (bottom).

With the setup and wave generation method unchanged, we swept the frequency regime from 100 to 20 000 Hz with a step size of 10 Hz and recorded the SPL data with and without the metasurface cage. Figure 5(c) shows the frequency responses and their SPL difference. A clear improvement in noise mitigation at extremely low frequencies (below 2300 Hz) can be observed. Essentially, this metasurface cage shows effective noise insulation starting at 300 Hz, and its insertion loss shows an overall decreasing trend until about 3230 Hz, where the global minimum of −60.362 dB is observed. Then the SPL difference floats around a mean value of −29.108 dB within the theoretical band gap. Even above the theoretical cutoff frequency, the cage remains effective, leading to an overall mean insertion loss of −23.407 dB from 100 to 20 000 Hz. The metasurface showed extraordinary noise mitigation in experiments, featuring 29-dB insertion loss across more than three octaves. The overall thickness of the metasurface is only $\lambda/10$ for 1500 Hz. Such a thin-profile, lightweight, and foldable structure can lead to significant advancement for broadband noise mitigation in water environments, especially for large construction applications.

As one may notice from Figs. 4(a) and 5(b), by the net effects of buoyancy, structural tension, and gravity, the unit cell is slightly deformed into a diamond shape rather than a perfect square lattice. This geometric morphing can be well characterized by the opening angle change of an individual unit cell. Following a similar eigenfrequency simulation as used to generate the band-structure diagram in Fig. 2(a), modal analysis is additionally conducted to quantitatively analyze the effect of induced opening angle variation. The results indicate that the plate wave band remains essentially unchanged, while the cutoff frequency increases with shrinking opening angle (i.e., resembling more a vertically hanging diamond). Hence, a reasonable amount of unit-cell deformation favors a wider attenuation band gap. See Appendix C for a more detailed analysis.

V. CONCLUSION AND DISCUSSION

In this work, we have developed a foldable and 3D-printable underwater acoustic metasurface that effectively mitigates broadband low-frequency waterborne noise. Inspired by the acoustic waveguide theory, the design leverages the special property of pressure-release boundaries which prevent wave propagation through the waveguide below a specific cutoff frequency. Because air has a much smaller characteristic impedance than water, air-entrained plates were employed as the sidewalls of each individual waveguide to create the pressure-release boundaries. To suppress undesired plate-shell vibrational modes that could open an extra low-frequency band and to ensure mechanical stability in the underwater environment, we selected a low-shear and lossy 3D-printable material (60A SuperElastic Resin) and incorporated internal wavy support structures within each plate. Furthermore, the metasurfaces are designed to be foldable for convenience of storage and transportation. The design is validated in both simulations and experiments.

The first prototype, referred to as the metasurface screen, demonstrated a strong noise attenuation capability with an average insertion loss of 20.7 dB within the theoretical band gap of 1.5 to 15 kHz, despite being affected by vibrational modes inherent to the lab-scale water tank at frequencies lower than 2300 Hz. An improved configuration, the metasurface cage, was developed to better decouple the sound source from the testing tank for more accurate measurements. This new setup shows a promising low-frequency noise reduction ability even below the band-gap onset and achieves a more consistent insertion loss of 29.1 dB across more than three octaves. Notably, the two configurations reached maximum insertion losses of 66.8 and 60.4 dB, respectively. Compared to the simulation results, the experimental measurements exhibit a lower attenuation performance. This discrepancy is partly attributed to the different metrics used for evaluation: simulations quantify the performance using transmission loss, which inherently accounts for additional scattering-induced energy redistribution that is not reflected in the insertion loss used for experiments.

In addition, the experimental environment introduces undesired transmission and interference, including additional reflection paths from the water–air interface and background broadband consistent noise from the laboratory HVAC system. Most importantly, electronic noise is introduced by the DAQ and hydrophone preamplifiers, making the background noise level around −70 dB compared with the signal level. For both experiments, we also observed consistent oscillatory behavior of the insertion loss curve. One primary explanation is the imperfect as-fabricated prototypes. Minor air bubble clusters attached to the structure can support additional high-frequency resonance modes, leading to certain low-amplitude oscillatory patterns in the high-frequency regime. Another reason is still due to environmental noise such as the laboratory HVAC system, intrinsic electronic noise, and power cable vibrations of the sound source.

However, due to the extremely high degree of freedom of the simulation system, full-scale frequency-domain simulations dealing with the water tank, acoustic source, and the entire metasurface simultaneously are not supported. Nevertheless, the idealized frequency-domain simulation of the metasurface can still reveal the overall acoustic behavior associated with the finite-size effects.

Attributed to its effectiveness, structural stability, broadband low-frequency performance, and ease of deployment, the proposed design shows strong potential for various real-life engineering applications. Bridge and offshore wind-farm constructions are often associated with intense low-frequency broadband underwater piling noise. We anticipate that deploying such a metasurface structure around piling sites will significantly mitigate such noise.

As a proof of concept, this study is still subject to further real-life improvements. Future research will be devoted to large-scale fabrication technology development and open-water field testing to validate practical performance. Additionally, the current literature has reported the speed of sound $c_{1\%} = 200$ m/s and mass density $\rho_e = 467.42$ kg/m^3 of void-embedded cement, which could achieve a relative acoustic impedance of only 0.062 [37]. Compared with the calculated surface impedance (approximately 0.08 at the maximum) of the air-entrained plate, such void-embedded cement has the potential to act as an acoustic soft boundary in harsh deep-water environments. Therefore, with continued advancements in composite materials design, where a rigid structural skeleton (e.g., concrete or metal) is integrated with compliant or voided inclusions, acoustic soft boundaries capable of withstanding high hydrostatic pressure are expected to become more feasible in the foreseeable future [38,39].

ACKNOWLEDGMENTS

This work was supported by start-up funding from Purdue University, US Department of Energy (DE-FOA-0003121), and Dow Chemical USA.

The various authors' contributions to this work were as follows: Y.Z., methodology, fabrication, experiment, data processing, visualization, writing; S.W., conceptualization, fabrication, experiment, data processing, writing; R.L., experiment; H.O., fabrication; Y.L., conceptualization; and J.L., conceptualization, methodology, experiment, data processing, writing.

Two of the authors, Y.Z. and J.L., have filed a patent for commercialization of the technology. All other authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

The data that support the findings of this article are openly available [40].

APPENDIX A: FREQUENCY SPECTRAL ANALYSIS OF METASURFACE WITH VARYING THICKNESS

To further clarify the performance limits of our design, additional frequency-domain simulations were conducted by varying the metasurface thickness (i.e., the air-entrained plate length). As shown in Fig. 6, increasing the longitudinal dimension mainly enhances the attenuation within the intended band-gap region, confirming the metasurface's ability to achieve stronger transmission reduction when additional decay distance is provided. For all four configurations, the strongest sound attenuation consistently occurs near the onset of the band gap, with quantitative evaluations of −76.48, −129.37, −189.69, and −233.59 dB, respectively, excluding abrupt reduction caused by local air-channel resonances.

As expected, the metasurface thickness has a limited influence on the plate-propagating wave regime and the transition stage. The average transmission loss of the plate wave modes remains approximately the same (−25.67 dB) for all cases. Additionally, the termination of the transition stage, or equivalently the onset of the band gap, occurs at the same frequency. This behavior is anticipated because the performance in this regime is primarily governed by the intrinsic mechanical properties of the air-entrained plates. Thickness variation only becomes significant if the plate is too short to provide sufficient decay space for the guided wave. Since the selected plate material exhibits significant structural damping, guided waves decay completely even in the shortest configuration, which corresponds to the cage design used in our experiments.

Moreover, the results show that the cutoff frequency is only weakly perturbed by metasurface thickness, which is consistent with the analytical understanding that f_c is predominantly determined by the in-plane waveguide profile for soft-boundary conditions. The minor oscillations near the cutoff frequency can be attributed to air-channel resonances.

APPENDIX B: INHERENT MODE ANALYSIS OF THE LAB-SCALE TESTING WATER TANK

In the first experiment using the metasurface screen, the SPL frequency responses with and without the structure exhibit nearly identical trends at very low frequencies (below 2300 Hz), with the metasurface case occasionally showing even worse responses. Although plate wave modes and the finite size of the plates may contribute to some unfavorable behaviors, the consistently similar responses indicate that a more fundamental property of the system likely dominates this phenomenon and masks the plate-related effects [Fig. 7(a)].

Besides the metasurface, another entity that possibly possesses intense low-frequency acoustic response is our lab-scale water tank ($1.22 \times 0.53 \times 0.45$ m^3). As a complex resonator, it supports a wide range of characteristic frequencies corresponding to the formation of standing-wave patterns inside the tank. Occasional mode matching could excite the inherent resonance modes of the water tank itself, resulting in massive power delivery through the glass wall. In COMSOL Multiphysics 6.2, an eigenfrequency simulation was conducted on a tank-water coupled model to validate its intrinsic resonance response. Figure 7(b) presents representative eigenmodes of the coupled system. For each panel (i–viii), the top part shows the displacement distribution of the glass walls (with purple indicating greater displacement), while the bottom part displays orthogonal slices of the total acoustic pressure within the water domain. The neighboring region of each

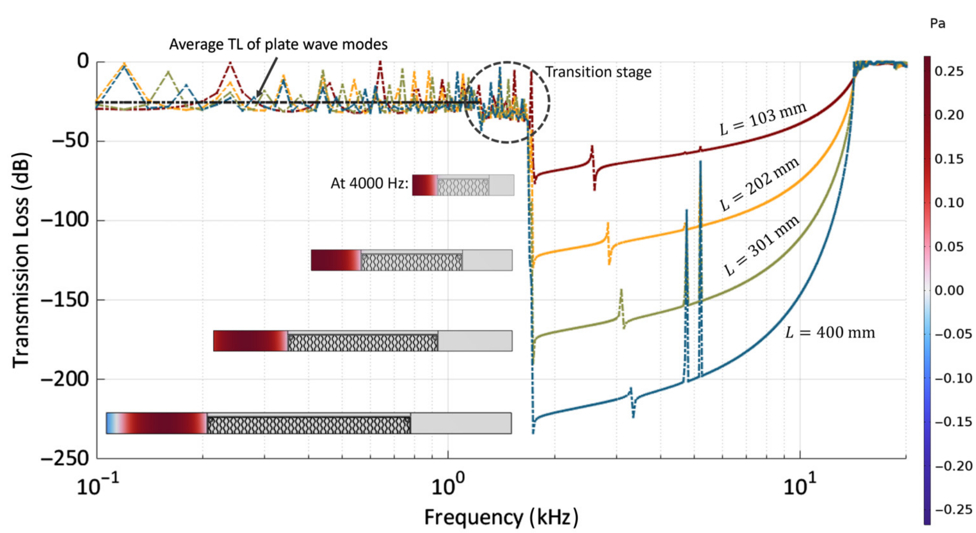

FIG. 6. Frequency-domain simulations with varying thickness of metasurface. Selected waveguide simulations illustrate the total acoustic pressure fields at 4000 Hz.

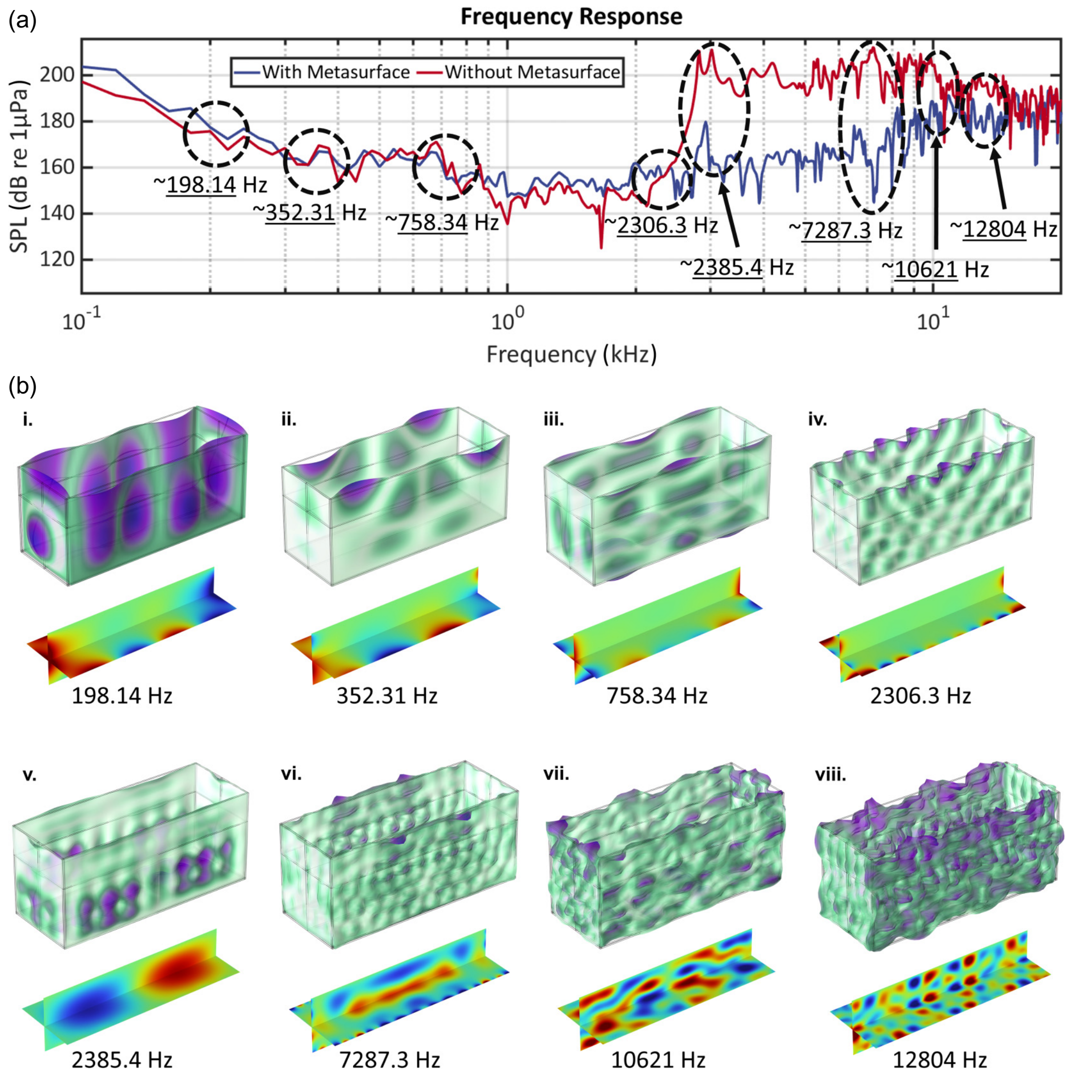

FIG. 7. (a) SPL frequency response curve with representative eigenfrequencies circled in black. (b) Representative eigenmode of the tank-water system. For each panel (i–viii), the top part displays tank glass deformation, and the bottom part displays the orthogonal slices of the total acoustic pressure field within the water domain.

eigenfrequency is also circled on the frequency response curves in Fig. 7(a).

Based on the effective operating range of the metasurface screen, the eigenmodes can be divided into two bands separated at approximately 2300 Hz. At the selected eigenfrequencies of 198.14, 352.31, 758.34, and 2306.3 Hz, the water tank exhibits vibrational modes. However, these modes appear to be highly contained to the tank itself. Although they have relatively strong coupling effects with the water near the sidewalls, their influence on the central region, where noise mitigation is most critical, is very limited. As a result, regardless of whether the metasurface is deployed, vibrations from the speaker on the incident side can still propagate to the receiving side and radiate secondary noise at these frequencies. This explains why, in the above frequency response curve, the effectiveness of the metasurface is ambiguous around these eigenfrequencies.

As frequency increases (or equivalently as the wavelength decreases), the water tank supports fewer resonant modes that can effectively couple energy into the water domain. At the representative eigenfrequency of 2385.4 Hz, the tank exhibits barely zero coupling with the water, while the acoustic field within the water displays a dipolelike pattern. This observation suggests that the metasurface is theoretically effective for noise mitigation at this frequency, which is supported by the significant SPL difference in the frequency response curve. Similarly, at the eigenfrequency of 7287.3 Hz, despite the trivial coupling at the sidewalls, a noticeable acoustic field still appears in the central region. As the frequency approaches the cutoff around 15 kHz, the two response curves tend to converge, again resulting in a reduced SPL difference even in the presence of the water tank.

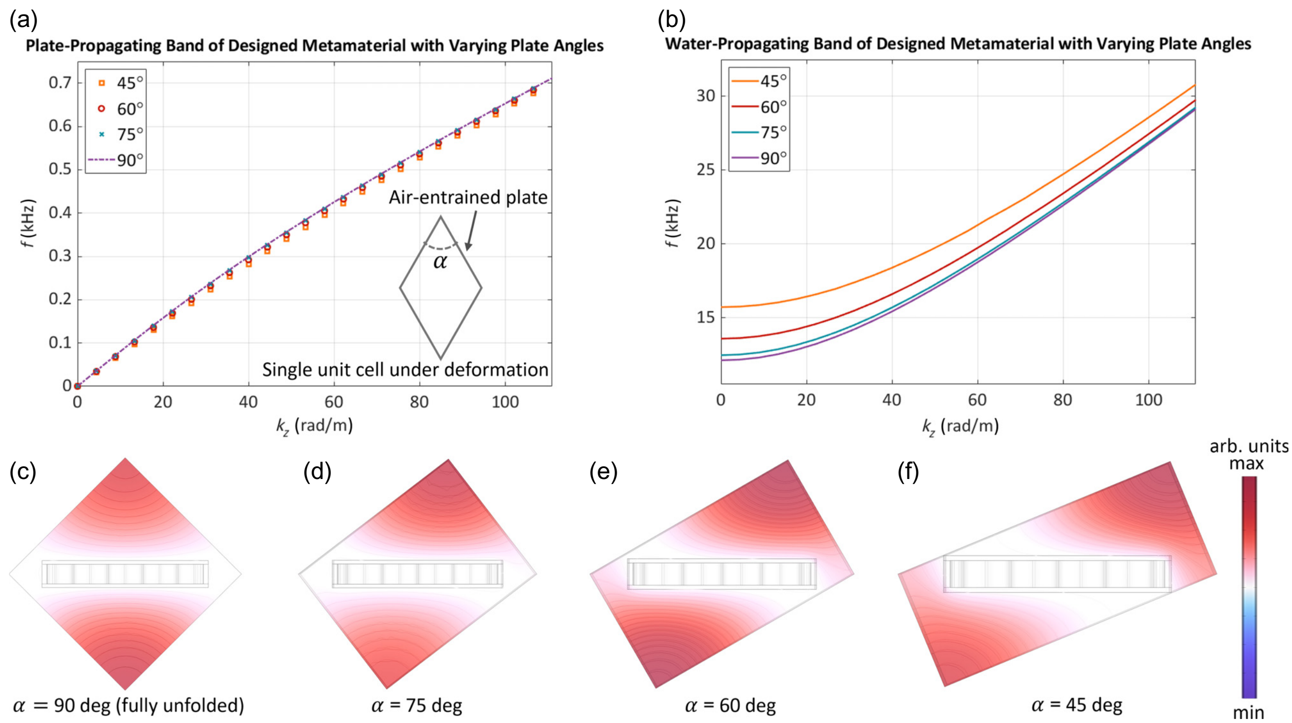

FIG. 8. Band structure with varying opening angles of the designed metasurface. (a) Plate-propagating bands. (b) Water-domain-propagating bands. (c)–(f) Representative water-propagating total pressure fields for angles from 90° to 45°, where a smaller angle corresponds to more vertical stretch of the metasurface.

Overall, this setup yields undesired performance in the low-frequency regime. One solution is to physically isolate the speaker from the glass walls, thereby significantly suppressing most of its inherent modes. This consideration led to our next design of a metasurface cage, which was explained in more detail in Sec. IV B.

APPENDIX C: ANALYSIS OF METASURFACE VARYING OPENING ANGLES

To quantitatively evaluate the influence of the opening angle, we performed eigenfrequency simulations for different folding states, with the corresponding band structures shown in Figs. 8(a) and 8(b). For computational efficiency, only a quarter of a single meta-atom profile is molded with proper symmetry boundary conditions.

The results indicate that, as the opening angle decreases, or equivalently as the metasurface is more vertically stretched, the cutoff frequency of the water-propagating band shifts to a higher value, resulting in a wider band gap. This demonstrates that reasonable tensile stretch is beneficial for broadening the noise-reduction bandwidth. Meanwhile, the plate-propagating bands remain nearly unchanged across all cases, since these modes are primarily determined by the mechanical properties of the air-entrained plates themselves.

It is noted that, for the 90° configuration, the cutoff frequency is $f_c = 12.13$ kHz, which slightly deviates from the result given in Fig. 2 of the main text, $f_c = 14.10$ kHz. This happens because the present simulation accommodates the “cage” design of individual plate, which uses slightly wider separation of 8 mm between two plates instead of 2 mm for the “screen” design due to the added thickness of the plate frame. Hence, a lower cutoff frequency is expected from the increased dimensions of the waveguide profile.

Finally, the smallest opening angle considered in our simulation is 45°, as further compact geometry would introduce excessive mechanical deformation and break the plate connections at the vertices.

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